Practical 1 – Weather Data Analysis using OpenWeatherMap API

Cell1- **What is the purpose?**

This cell imports the libraries required to collect, store, analyse and visualize weather data.

Library	**Purpose**
os	Performs operating-system and file-related operations.
requests	Sends HTTP requests to the weather API.
pandas	Stores and handles weather data in table format.
matplotlib	Creates graphs.
seaborn	Creates statistical visualizations.

Last two statements:

- figure.figsize = (10, 5) sets the default graph size.
- axes.grid = True displays grid lines on graphs.

Expected output

No output. This cell only imports libraries and sets graph properties.

Viva questions

1. What is an API? An API allows two software applications to communicate with each other.
2. What is OpenWeatherMap? It is a weather service that provides weather information through an API.
3. Why do we use Requests? To send HTTP requests and retrieve data from the API.
4. Why do we use Pandas? To store and manipulate data in tabular form.

   Cell 2 – Fetching weather data from the API

   This is the main data-collection cell.

   Purpose: It fetches live weather information for five cities: Pune, Mumbai, Delhi, Bangalore and Chennai.

   The cell performs these tasks:

1. Stores the OpenWeatherMap API key.
2. Creates a list of five cities.
3. Sends a request for each city.
4. Receives weather information in JSON format.
5. Extracts temperature, humidity, wind speed, precipitation and geographical coordinates.
6. Stores the information in a Pandas DataFrame.
7. Handles missing values.

Important concepts

- API key: A unique key used to authenticate API requests.
- HTTP request: A request sent to a server to obtain information.
- JSON: A format used to exchange structured data.
- DataFrame: A table containing rows and columns.
- Missing values: Data that is unavailable or empty.

Cell 3 – Statistical Analysis

Purpose: Calculate basic statistics from the collected weather data.

The cell calculates:

Statistic	**Meaning**	**Journal result**
Average temperature	Mean temperature of all cities	32.802°C
Maximum temperature	Highest temperature	35.24°C
Minimum temperature	Lowest temperature	30.80°C
Average humidity	Mean humidity	45.2%
Average wind speed	Mean wind speed	3.182

Important concepts

- Mean: Average value.
- Maximum: Largest value.
- Minimum: Smallest value.

Cell 4 – Temperature Visualization

Purpose: Compare the temperatures of different cities using a bar graph.

- X-axis: City names.
- Y-axis: Temperature in Celsius.
- Each bar represents one city's temperature.

Important: Delhi has the highest temperature, and Bangalore has the lowest in the recorded dataset.

Viva: Why do we use a bar graph?

To compare numerical values between different categories.

Cell 5 – Humidity Visualization-

Purpose: Compare humidity levels across the five cities.

- X-axis: City names.
- Y-axis: Humidity percentage.
- Each bar represents the humidity of one city.

  In the journal data, Mumbai has the highest humidity (64%), while Pune has the lowest (25%).

  Viva: What is humidity?

  Humidity is the amount of water vapour present in the air.

  Cell 6 – Temperature vs Humidity-

  Purpose: Study the relationship between temperature and humidity using a scatter plot.

- X-axis: Temperature (°C).
- Y-axis: Humidity (%).
- Each dot represents one city.

  What is a scatter plot?

  A scatter plot displays the relationship between two numerical variables using points.

  Viva: Why do we use a scatter plot?

  To identify patterns, relationships and possible trends between two numerical variables.

  Cell 7 – Correlation Heatmap-

  Purpose: Find the correlation between weather variables.

  The notebook calculates correlation for:

- Temperature
- Humidity
- Wind speed
- Precipitation

  What is correlation?

  Correlation measures how two variables are related.

Correlation value	**Meaning**
+1	Perfect positive correlation
0	No linear correlation
-1	Perfect negative correlation

  What is a heatmap?

  A heatmap represents numerical values using different colours. The annot=True setting displays the correlation values inside the cells.

  Viva: Why do we use a heatmap?

To easily visualize the strength and direction of relationships between multiple variables.

Cell 8 – Geographic Visualization

Purpose: Display the geographical positions of the five cities using latitude and longitude.

- X-axis: Longitude.
- Y-axis: Latitude.
- Dots: Locations of cities.
- City labels: Names of the cities.

Important concepts:

- Latitude: Measures how far a location is north or south of the Equator.
- Longitude: Measures how far a location is east or west of the Prime Meridian.

Viva: Why do we use latitude and longitude?

To identify and visualize the geographical locations of places.

1. What is the aim of this practical?
To collect weather data using the OpenWeatherMap API and analyse it through statistics and visualizations.

2. Which API is used?
OpenWeatherMap API.

3. Which cities are considered?
Pune, Mumbai, Delhi, Bangalore and Chennai.

4. What is JSON?
JavaScript Object Notation, a format used to exchange data.

5. What is a DataFrame?
A two-dimensional tabular data structure provided by Pandas.

6. What is data visualization?
Representing data graphically to understand it easily.

7. What is correlation?
A statistical measure that describes the relationship between two variables.

8. What is the difference between a bar graph and a scatter plot?
A bar graph compares categories, while a scatter plot shows the relationship between numerical variables.

9. What are the uses of an API?
APIs allow applications to communicate and exchange data.

10. What is precipitation?
Water falling from the atmosphere in the form of rain, snow or other forms.

How to explain the complete practical to your examiner

You can say:

"This practical is based on Weather Data Analysis using the OpenWeatherMap API. First, we import the required Python libraries. Then we connect to the API and collect weather information for five cities: Pune, Mumbai, Delhi, Bangalore and Chennai. We store the collected information in a Pandas DataFrame and handle missing values. After that, we calculate average, maximum and minimum temperature, average humidity and wind speed. Finally, we create bar graphs, a scatter plot, a correlation heatmap and a geographical visualization to understand and compare the weather conditions."

Practical 2

DMV Practical 2 – Telecom Customer Churn Data Preprocessing

This practical is about cleaning and preparing telecom customer data so that it can be used for machine learning.

First, understand the practical

What is customer churn?

Customer churn means a customer stops using a company's services.

For example, if a person stops using their current telecom network and switches to another company, that person is considered to have churned.

Aim: To clean and preprocess the telecom customer dataset by handling missing values, standardizing data, handling outliers, encoding categorical variables and scaling numerical features.

Cell 1 – Importing libraries

What is the purpose?

This cell imports the libraries required for data cleaning and preparation.

Library	**Purpose**
Pandas	Handles datasets in table format.
NumPy	Performs numerical operations.
train_test_split	Divides data into training and testing sets.
StandardScaler	Standardizes numerical data.

Important concepts

1. Pandas: Used to read, modify and analyse datasets.

2. NumPy: Used for mathematical and numerical operations.

3. train_test_split: Divides the dataset into two parts:

- Training data – used to train a model.
- Testing data – used to test a model.

4. StandardScaler: Brings numerical features to a comparable scale.

Expected output

No output. This cell only imports libraries.

Viva questions

Q1. What is Pandas?

Pandas is a Python library used for data manipulation and analysis.

Q2. What is NumPy?

NumPy is a Python library used for numerical calculations.

Q3. Why do we use StandardScaler?

To standardize numerical data so that features have a comparable scale.

Cell 2 – Loading the dataset

What is the purpose?

This cell loads the telecom customer dataset into Python.

Let's understand the important terms.

1. pd.read_csv()

Reads a CSV file and loads it into a Pandas DataFrame.

2. df

A variable that stores our dataset.

3. df.head()

Displays the first five rows of the dataset.

4. df.shape

Displays the number of rows and columns.

Output from your journal

Dataset shape

7043 × 21

This means:

- 7043 rows = 7043 customer records.
- 21 columns = 21 attributes or features.

Some columns include:

Column	Meaning
customerID	Unique customer identification
gender	Customer's gender

SeniorCitizen	Whether customer is a senior citizen
Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months with the company
PhoneService	Whether phone service is used
MonthlyCharges	Monthly bill
TotalCharges	Total charges
Churn	Whether customer left the company

Viva questions

Q1. What is a CSV file?

CSV stands for Comma-Separated Values. It stores data in rows and columns.

Q2. What is a DataFrame?

A DataFrame is a two-dimensional data structure containing rows and columns.

Q3. What does head() do?

It displays the first five rows by default.

Q4. What does shape return?

It returns the number of rows and columns.

Cell 3 – Dataset Information

What is the purpose?

This cell provides detailed information about the dataset.

It displays:

1. Number of rows.
2. Number of columns.
3. Column names.
4. Non-null values.
5. Data types.

Important concepts

1. Non-null values: Values that are not missing.

2. Data types: Types of values stored in columns.

Common data types:

Data type	Meaning	Example

int64	Integer values	25
float64	Decimal values	64.75
object	Text or categorical values	Male, Female

In this dataset, columns such as gender, Partner and Churn are categorical, while tenure and MonthlyCharges are numerical.

Expected output

The notebook shows 7043 entries and 21 columns, along with their data types and non-null counts.

Viva questions

Q1. What is df.info()?

It provides information about the dataset, including column names, data types and non-null counts.

Q2. What is the difference between int and float?

Int stores whole numbers, while float stores decimal numbers.

Q3. What is an object data type?

It is commonly used for storing text and categorical values.

Cell 4 – Dataset Exploration

Your journal moves from the dataset information section to the heading Explore the dataset. The exported report does not clearly show a separate executed code cell numbered 4, so I won't invent its code.

The next visible executed cell is Cell 5.

Cell 5 – Summary Statistics

Summary Statistics**What is the purpose?**

This cell calculates statistical information about the numerical columns in the dataset.

It displays:

Term	**Meaning**
Count	Number of values
Mean	Average value
Std	Standard deviation
Min	Minimum value
25%	First quartile
50%	Median

75%	Third quartile
Max	Maximum value

Important output from your journal

Column	Mean	Minimum	Maximum
SeniorCitizen	0.162	0	1
Tenure	32.37	0	72
MonthlyCharges	64.76	18.25	118.75

How to understand this output?

- Average customer tenure is approximately 32 months.
- Average monthly charges are approximately 64.76.
- Maximum tenure is 72 months.

Viva questions

Q1. What is descriptive statistics?

It is the process of summarizing and understanding data using statistical values.

Q2. What is mean?

Mean is the average of all values.

Q3. What is standard deviation?

It measures how much the values are spread around the mean.

Q4. What does describe() do?

It displays summary statistics of numerical columns.

Cell 6 – Checking Missing Values

What is the purpose?

This cell checks whether the dataset contains missing or empty values.

Understanding the code

- df.isnull() checks each cell for missing values.
- It returns True if a value is missing and False otherwise.
- .sum() counts the missing values in each column.

Output

Your journal shows zero missing values across the dataset columns.

Result

No missing values were found.

Viva questions

Q1. What are missing values?

Missing values are empty or unavailable values in a dataset.

Q2. Why do we check missing values?

Because missing data can affect analysis and machine learning results.

Q3. What is isnull()?

It checks whether values in a dataset are missing.

Cell 7 – Checking Duplicate Records

Cell 8 **Cell 8: Fill Missing Values**

Purpose: This cell is used to handle missing values in the dataset.

In this cell, missing values in numerical columns are replaced with the median of that column, while missing values in categorical columns are replaced with the mode (most frequently occurring value). Median is used for numerical data because it is less affected by outliers, whereas mode is suitable for categorical data because it represents the most common category.

After filling the missing values, the program displays the missing-value count for each column using df.isnull().sum(). If all missing values are handled successfully, the output will show 0 for every column.

Remember for viva:

- Numerical data → Median
- Categorical data → Mode
- fillna() → Fills missing values.
- isnull().sum() → Counts missing values.

In simple words: This cell cleans the dataset by filling empty numerical and categorical values with suitable replacements.

Cell 9 – Standardizing Text Values

What is the purpose?

This cell standardizes text values in the dataset.

For example:

Before	**After**
Male	male
Female	female
YES	yes
No	no

Understanding important terms

1. select_dtypes(include='object')

Selects all columns containing text or object data.

2. strip()

Removes extra spaces from the beginning and end of text.

3. lower()

Converts all text into lowercase.

Why do we do this?

Because Yes, YES and yes should be treated consistently instead of being treated as different categories.

Expected output

Categorical values standardized.

Viva questions

Q1. What is data standardization?

It means making data values consistent in format.

Q2. What does strip() do?

It removes unnecessary spaces from the beginning and end of text.

Q3. Why use lower()?

To convert text into lowercase and maintain consistency.

Cell 10 – Converting Data Types

What is the purpose?

This cell ensures that the columns have appropriate data types.

Understanding the code

1. astype(str)

Converts customer ID into a string because it is an identifier, not a numerical measurement.

2. pd.to_numeric()

Converts the selected columns into numerical values.

3. errors="coerce"

If a value cannot be converted into a number, it is converted into NaN (missing value).

4. df.dtypes

Displays the data type of every column.

Output

The notebook displays the updated data types. For example:

- customerID → object
- tenure → int64
- MonthlyCharges → float64

Viva questions

Q1. What is type conversion?

Changing data from one data type to another.

Q2. Why do we convert customerID into a string?

Because it is an identifier and should not be used as a numerical measurement.

Q3. What is NaN?

NaN means Not a Number and is commonly used to represent missing numerical values.

Cell 11 – Handling Outliers using IQR

What is an outlier?

An outlier is a value that is unusually high or unusually low compared to other values.

For example, if most customers have monthly charges between 20 and 100, a value much higher than this range may be an outlier.

What is IQR?

IQR stands for Interquartile Range.

It measures the spread of the middle 50% of the data.

Formula:

Iqr =q3-q1

Where:

- Q1 = 25th percentile.
- Q3 = 75th percentile.

How are outliers identified?

Lower- q1-1.5(iqr)

Upper= Q3+1.5(IQR)

What is the purpose?

This cell indicates that the data-cleaning stage is complete and the dataset is ready for the next preparation steps.

The journal explicitly mentions that no extra feature engineering is added beyond the available dataset fields.

Output

Dataset ready for model preparation.

Viva question

Q. What is feature engineering?

Feature engineering is the process of creating or modifying features to make data more useful for machine learning.

Cell 13 – Encoding Categorical Data

Values outside these limits are treated as outliers.

What does clip() do?

It replaces values outside the limits with the nearest limit.

For example, if the upper limit is 100 and a value is 150, clipping changes it to 100.

Why is SeniorCitizen skipped?

Because it is a binary indicator with values 0 and 1, not a continuous measurement.

Expected output

Outliers handled using IQR method.

Viva questions

Q1. What is an outlier?

An unusually high or low value in a dataset.

Q2. What is IQR?

Interquartile Range, calculated as Q3 minus Q1.

Q3. Why do we handle outliers?

To reduce the influence of extreme values on data analysis and modelling.

Q4. What does clip do?

It limits values to a specified lower and upper range.

Cell 12 – Preparing Dataset for Modelling

What is the purpose?

This cell indicates that the data-cleaning stage is complete and the dataset is ready for the next preparation steps.

The journal explicitly mentions that no extra feature engineering is added beyond the available dataset fields.

Output

Dataset ready for model preparation.

Viva question

Q. What is feature engineering?

Feature engineering is the process of creating or modifying features to make data more useful for machine learning.

Cell 13 – Encoding Categorical Data

What is the purpose?

This cell converts categorical data into numerical form so that it can be used by machine learning algorithms.

Understanding the code

1. drop()

Removes the customerID and Churn columns from the feature dataset.

- Customer ID is only an identifier.
- Churn is the target variable that we want to predict, so it is kept separate.

2. pd.get_dummies()

Converts categorical columns into numerical columns using one-hot encoding.

Example:

Gender	Gender_male
female	0
male	1
female	0

3. drop_first=True

Drops the first category to avoid unnecessary duplicate information.

4. dtype=int

Stores encoded values as integers (0 and 1).

Output

Encoded dataset shape

(7043, 30)

This means the encoded dataset contains 7043 rows and 30 columns.

Viva questions

Q1. What is encoding?

Encoding converts categorical data into numerical form.

Q2. Why is encoding required?

Most machine learning algorithms require numerical input.

Q3. What is one-hot encoding?

It represents categories using separate binary columns.

Q4. What is get_dummies()?

It is a Pandas function used to convert categorical variables into dummy variables.

Cell 14 – Separating Target Variable

What is the purpose? What is the purpose?

This cell converts the Churn column into binary numerical values and separates input features from the target.

Understanding the code

1. value_counts()

Counts how many times each value appears in the Churn column.

Output from your journal

Churn	Count
No	5174
Yes	1869

2. map({"yes": 1, "no": 0})

Converts:

- Yes → 1
- No → 0

3. X and y

This is an important machine learning concept.

- X = Input features used for prediction.
- y = Target variable we want to predict.

In this practical, the target is whether the customer leaves the telecom company.

Viva questions

Q1. What is a target variable?

It is the output that a machine learning model tries to predict.

Q2. What is X?

X contains the input features.

Q3. What is y?

y contains the target or output variable.

Q4. What is binary encoding?

Representing two categories using 0 and 1.

Cell 15 – Feature Scaling using StandardScaler

This cell converts the Churn column into binary numerical values and separates input features from the target.

Understanding the code

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Representing two categories using 0 and 1.

Cell 15 – Feature Scaling using StandardScaler

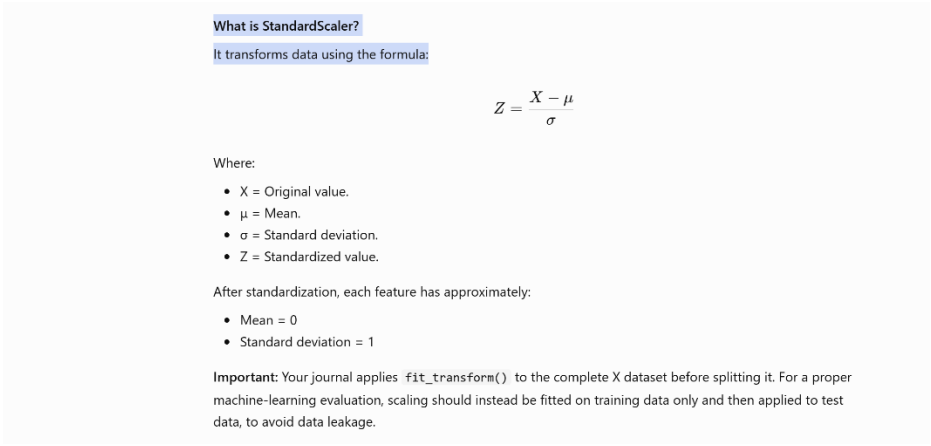
What is the purpose?

This cell standardizes the input features so that they are on a comparable scale.

For example, tenure may range from 0 to 72, while monthly charges may range from 18 to 119. Different scales can affect some machine learning algorithms.

What is StandardScaler?

It transforms data using the formula:



What is StandardScaler?

It transforms data using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = Original value.
- μ = Mean.
- σ = Standard deviation.
- Z = Standardized value.

After standardization, each feature has approximately:

- Mean = 0
- Standard deviation = 1

Important: Your journal applies `fit_transform()` to the complete X dataset before splitting it. For a proper machine-learning evaluation, scaling should instead be fitted on training data only and then applied to test data, to avoid data leakage.

Viva questions

Q1. What is feature scaling?

It is the process of bringing numerical features to a comparable scale.

Q2. What is StandardScaler?

A method that standardizes data using mean and standard deviation.

Q3. What is the difference between normalization and standardization?

Normalization usually scales values to a fixed range, while standardization transforms values based on mean and standard deviation.

Cell 16 – Train-Test Split

What is the purpose?

This cell divides the prepared dataset into training and testing sets.

Training data: Used to train a machine learning model.

Testing data: Used to evaluate the model on unseen data.

Understanding important parameters

- test_size=0.20: 20% of data is used for testing.
- random_state=42: Helps make the split reproducible.

- **Viva questions**
- Q1. What is train-test split?
- It divides data into training and testing sets.
- Q2. Why do we split data?
- To evaluate how well a model performs on unseen data.
- Q3. What does test_size=0.20 mean?
- 20% of the dataset is reserved for testing.
- Q4. What is random_state?
- It ensures reproducibility of the random split.

Cell 17 – Exporting the Cleaned Dataset

What is the purpose?

This cell saves the cleaned and encoded datasets as CSV files.

Two files are generated:

1. Telecom_Customer_Churn_Cleaned.csv – stores the cleaned dataset.
2. Telecom_Customer_Churn_Encoded.csv – stores the encoded dataset with the Churn target.

What does to_csv() do?

It exports a Pandas DataFrame into a CSV file.

What does index=False mean?

It prevents Pandas from saving the row index as an additional column.

Expected output

- Cleaned dataset exported successfully!
- Encoded dataset exported successfully!

Viva questions

Q1. What is to_csv()?

It saves a DataFrame into a CSV file.

Q2. Why do we use index=False?

To avoid saving the DataFrame index as an extra column.

Important viva questions for the complete practical

1. What is data preprocessing?
It is the process of cleaning and preparing raw data before machine learning.

2. What is customer churn?
It means a customer stops using a company's services.

3. Which dataset is used?
Telecom Customer Churn dataset.

4. How many records are present?
7043 records.

5. How many columns are present initially?
21 columns.

6. What is data cleaning?
Removing errors, inconsistencies and unwanted values from data.

7. What are categorical variables?
Variables that represent categories, such as gender and contract type.

8. What are numerical variables?
Variables containing numerical values, such as tenure and monthly charges.

9. What is IQR?
Interquartile Range, calculated as Q3 minus Q1.

10. What is one-hot encoding?
Converting categories into separate binary columns.

11. What is feature scaling?
Bringing numerical features to a comparable scale.

12. What is the final training and testing split?
5634 training records and 1409 testing records.

How to explain this practical to your examiner

"This practical is based on Telecom Customer Churn Data Preprocessing. First, we import the required libraries and load the telecom dataset containing 7043 records and 21 columns. We explore the dataset, check its information, summary statistics and missing values. Then we standardize text values, convert data types and handle outliers using the IQR method. After that, we encode categorical variables, separate the input features and target variable, apply StandardScaler and divide the data into training and testing sets. Finally, we export the cleaned and encoded datasets into CSV files."

Quick revision – Remember these important outputs

Step	**Result**
Original dataset	7043 rows, 21 columns
Missing values	0 in all columns
Encoded dataset	7043 rows, 30 columns
Churn = No	5174
Churn = Yes	1869
Training data	5634 records
Testing data	1409 records
Exported files	2 CSV files